

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
 BURP 3rd Year 2nd Term Regular Examination, 2016.
CE 3207
 (Elements of Civil Engineering Structures)

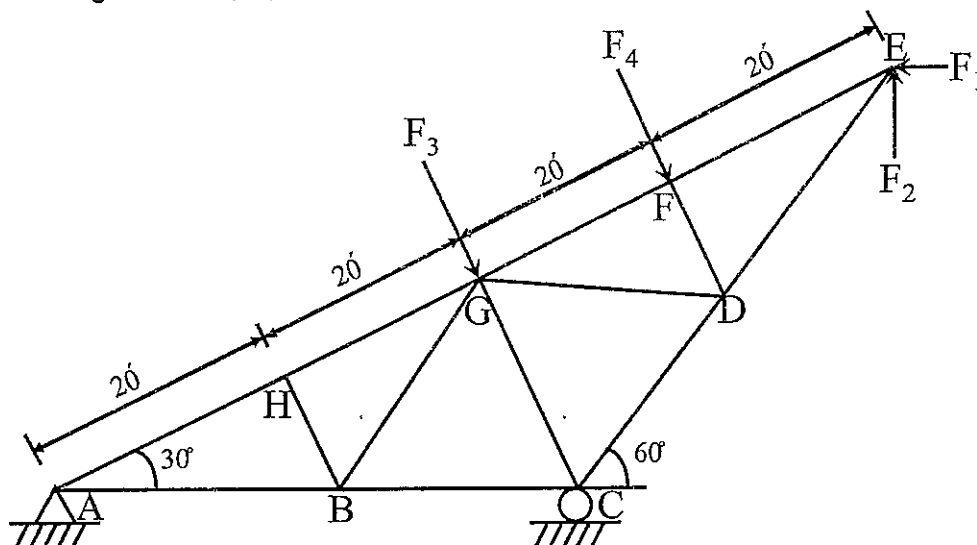
Full Marks: 210

Time: 3.0 hrs

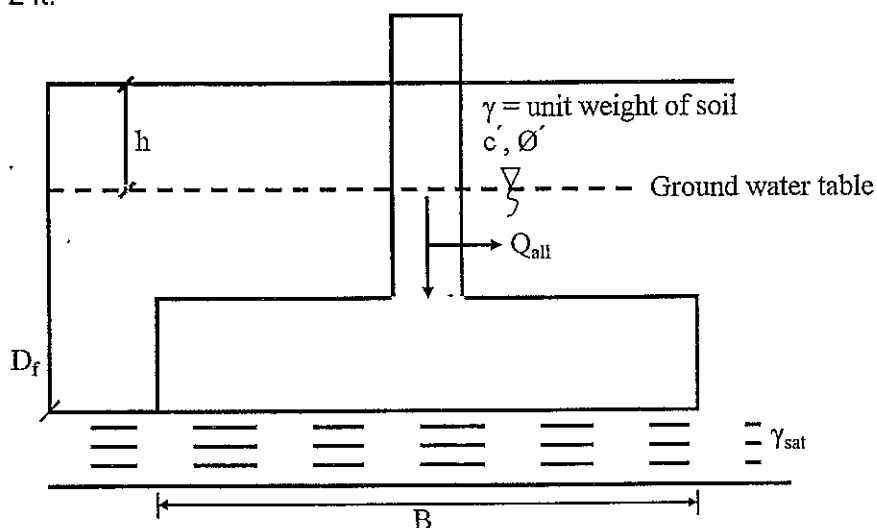
- N.B.: i) Answer any **three** questions from each section in separate script
 ii) Figures in the right margin indicate marks
 iii) Assume any reasonable value for missing data

Section - A

1. (a) What is meant by foundation? Classify the general types of foundation with figure. (13)
 (b) Write down the considerations which are required for different types of foundation in clay and sand. (12)
 (c) What type of problems may arise in foundation for different types of soil? What are the possible solutions to these problems? (10)
2. (a) What is truss? Explain different types of truss with neat sketches. (10)
 (b) Compute the external reactions and loads on the members FD, HG and BC as shown in figure. Here, $F_1 = 2000$ lb, $F_2 = 3000$ lb, $F_3 = 4000$ lb and $F_4 = 5000$ lb. (25)



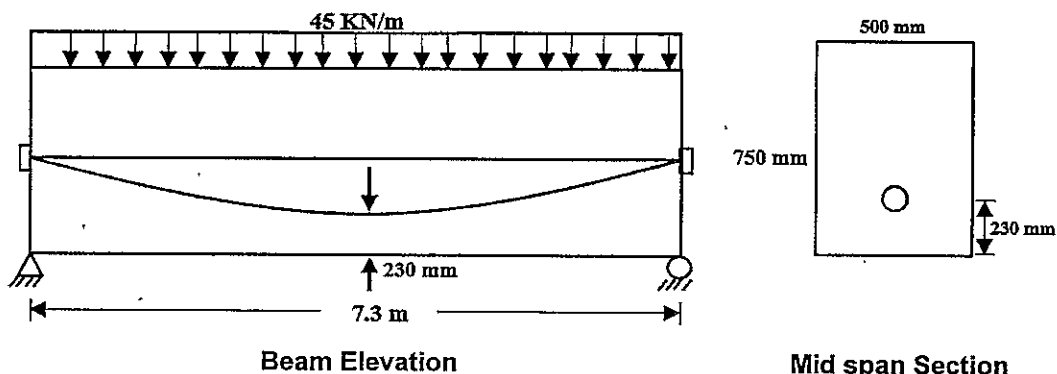
3. (a) Deduce the Terzaghi's bearing capacity equation. (12)
 (b) Explain the test procedure of "Plate Load Test" to determine the bearing capacity of soil. (10)
 (c) A square footing as shown in figure below. Determine the gross allowable load, Q_{all} , that the footing can carry. Use Terzaghi's equation for general shear failure ($F_s = 3$). Given $r = 105$ lb/ft³, $r_{sat} = 118$ lb/ft³, $c = 10$ kPa, $\phi = 35^\circ$, $B = 5$ ft, $D_f = 4$ ft and $h = 2$ ft. (13)



4. (a) What is meant by consolidation and settlement? Draw and explain the contact pressure and settlement profile of flexible and rigid foundation in clay and sand. (12)
- (b) Define normally consolidated and over consolidated soil. Explain the variation of total stress, pore water pressure and effective stress in a clay layer drained at top and bottom, subjected to stress $\Delta\sigma$. (13)
- (c) What are the effects of water table on soil bearing capacity? (10)

Section – B

5. (a) What is meant by reinforced concrete? Why is mild steel used in reinforced concrete? (06)
- (b) Write down the advantages of reinforced concrete over the plain concrete. (07)
- (c) Discuss different types of failure crack in RCC beam with neat sketches. (07)
- (d) A simply supported rectangular beam with a span of 18 ft, supports a uniform live load of 960 Plf and dead load of a wall weighing 350 Plf in addition to its own weight, with $f'_c = 3000$ psi; and $f_y = 60000$ psi. Determine the required steel area. Draw the x-section and long section of the beam. (15)
6. (a) Differentiate between- (09)
 - (i) WSD and USD Method.
 - (ii) False T beam and Real T beam.
 - (iii) One way and Two way slab.
- (b) What are the basic assumptions of reinforced concrete design? (07)
- (c) Draw the stress-strain diagram of mild steel and concrete. (04)
- (d) Discuss different types of prestressed concrete in brief. (15)
7. (a) What is column? Write down the functions of column. (06)
- (b) Differentiate between- (09)
 - (i) Short column and slender column.
 - (ii) Composite column and combination column.
 - (iii) Tied column and spiral column.
- (c) Design a square tied column to support an axial live load of 170 k and dead load of 150 k. Initially assume that 2% longitudinal steel is desired. Assume, $f'_c = 4000$ psi and $f_y = 60000$ psi. (20)
8. (a) What is prestressed concrete? Write down the advantages and disadvantages of prestressed concrete over the reinforced concrete? (08)
- (b) Write down the concrete strength requirements of prestressed concrete. (06)
- (c) Why high tensile steel is used in prestressed concrete? (06)
- (d) A prestressed concrete rectangular beam 500 mm by 750 mm has a simple span of 7.30 m and is loaded by a uniform load of 45 KN/m including its own weight. The parabolically curved prestressing tendon is located as shown in figure below and produces an effective prestress of 1620 KN. Compute the extreme fiber stresses at mid span by the method of load balancing. (15)



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2016
CE 3261
(Water Resources Planning)

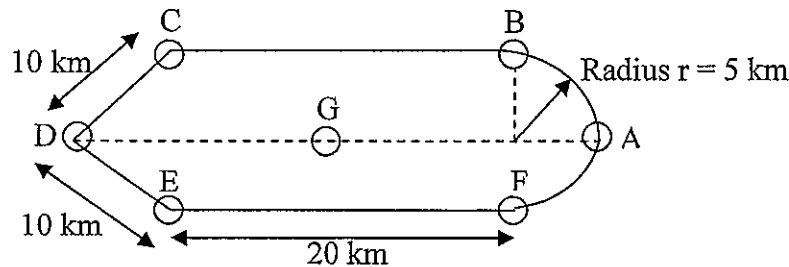
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Plain graph paper will be supplied
iv) Assume Reasonable values for any missing data

Section – A

1. (a) What is hydrology? Describe the hydrologic cycle. (11)
(b) What do you mean by probable maximum precipitation? Find the mean precipitation for the area sketched below by Thiessen polygon method. (12)



Station	A	B	C	D	E	F	G
Rainfall (cm)	6.8	6.4	4.8	5.4	6.9	5.5	7.5

- (c) A catchment has six rain-gauge stations. In a year the annual rainfall recorded by the gauges are as follows- (12)

Station	A	B	C	D	E	F
Rainfall (cm)	82.6	102.9	180.3	110.3	98.8	136.7

For a 10% error in the estimation of the mean rainfall, calculate the optimum number of stations in the catchment.

2. (a) What is evaporation? Describe the energy budget method for determination of the lake evaporation. (10)
(b) Differentiate between – (12)
i) Φ - index and w - index
ii) Base flow and direct runoff
iii) Hydrograph and hyetograph
(c) Rainfall of magnitude 3.8 cm and 2.8 cm occurring on two consecutive 4-h durations on a catchment of area 27 km² produced the following hydrograph of flow at the outlet of the catchment. Estimate the rainfall excess and Φ - index. (13)

Time from start of rainfall (h)	-6	0	6	12	18	24	30	36	42	48	54	60	66
Observed flow (m ³ /s)	6	5	13	26	21	16	12	9	7	5	5	4.5	4.5

3. (a) Draw a typical hydrograph and show the components of hydrograph. (07)
(b) Explain the area velocity method of measuring discharge of a stream. (10)
(c) Given the ordinates of a 4-h unit hydrograph as below derive the ordinates of a 12-h unit hydrograph for the same catchment by S-curve method. (18)

Time (h)	0	4	8	12	16	20	24	28	32	36	40	44
Ordinate of 4-h UH	0	20	80	130	150	130	90	52	27	15	5	0

4. (a) For the confined flow into a well at the edge of the zone of influence show that, (12)

$$Q = \frac{2\pi T S_w}{\ln \frac{R}{r_w}}$$

- (b) Differentiate between – (09)
i) Specific yield and specific retention;
ii) Aquiclude and aquifuge;
iii) Water table aquifer and artesian aquifer.

- (c) A 30 cm well completely penetrates an unconfined aquifer of saturated depth 40 m. (14)
After a long period of pumping at a steady rate of 1500 lpm, the drawdown in two observation wells located 25 and 75 m from the pumping well were found to be 3.5 and 2.0 m respectively. Determine the transmissibility of the aquifer. What is the drawdown at the pumping well?

Section – B

5. (a) What do you mean by delta of a crop? Derive a relationship between duty and delta for a given base period. (07)
(b) Write short notes on- (15)
(i) Furrow irrigation method; (ii) Basin flooding; (iii) Drip irrigation method.
(c) A stream of 150 liter/sec was diverted from a canal and 110 liter/sec was diverted to the field. An area of 16,000 m² was irrigated in 8 hrs. The effective depth of root zone was 1.8 m. The runoff loss in the field was 400 cu.m. The depth of water penetration varied linearly from 1.9 m at the head end of the field to 1.2 m at the tail end. Available moisture holding capacity of the soil is 20 cm/m depth of soil. Determine the water conveyance efficiency, water application efficiency, water storage efficiency, and water distribution efficiency. Irrigation was started at a moisture extraction level of 40% of the available moisture. (13)
6. (a) What is flood? Write down the causes of flood in Bangladesh. (12)
(b) Illustrate the importance of design flood with its definition. Discuss Dicken's Formula, Ryve's Formula, and Rational Formula for estimating design flood. (13)
(c) On the basis of isopluvial maps the 50 year 24 hr maximum rainfall at KUET is found to be 20 cm. Determine the probability of 24 hr rainfall of magnitude equal to or greater than 20 cm occurring at KUET. (10)
i) at least once in 20 successive years;
ii) two times in 20 successive years;
iii) once in 20 successive years.
7. (a) Define return period. Describe briefly the IDF curve. (07)
(b) Write short notes on- (16)
(i) Groyne, (ii) Vegetation; (iii) Guide banks; (iv) Artificial cut-offs.
(c) For a river, the estimated flood peaks for two periods by the use of Gumbel's method are given below. (12)

Return period (yr)	Peak flood (m ³ /s)
100	500
50	465

What flood discharge in this river will have a return period of 1000 years?

8. (a) What do you mean by river training works? What are the objectives of river training works? (07)
(b) What is meant by dredging? Briefly describe different types of dredgers. (10)
(c) The Flood Frequency Study on a river has shown that during 50 years 125 floods occurred. The magnitude of each flood was known from which the number of floods falling between the two limits of flood peak were counted and tabulated in the table below. (18)

Flood peak limit (m ³ /s)	No. of floods
650 - 709	1
710 - 779	5
780 - 844	30
845 - 914	22
915 - 989	20

Flood Peak limit (m ³ /s)	No. of floods
990 - 1059	20
1060 - 1129	11
1130 - 1269	10
1270 - 1414	5
1415 - 1699	1

Evaluate the magnitude of flood having a frequency of

- (i) 10 years (ii) 30 years and (iii) 90 years.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2016
URP 3231
(Transportation Planning)

Full Marks: 210

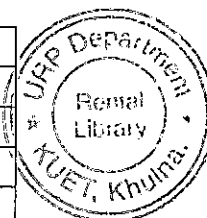
Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What are the selection perspectives of transportation modes in urban setting? (08)
Discuss about the national perspective.
- (b) Mention the reasons to support walking. What are the components of walking (10)
in urban transportation system?
- (c) How the pedestrian circulation in spaces is determined? Write down the (07)
factors that influence the walkability.
- (d) Find out the area capacity for pedestrian circulation of a Mezzanine of Metro (10)
station from the following information:

Types of Movement (per hour)	Number of Persons	Time
Stair to turnstiles	3,200	15 sec
Stairs to token booth to turnstiles	600	30 sec
Stairs to newsstand to turnstiles	200	60 sec
Turnstiles to newsstand to stairs	100	45 sec



Assume that the area of the Mezzanine is 1,500 sq.ft including an unused area of 350 sq.ft and expected Level of Service (LOS) is D.

2. (a) Make a comparative overview among Curb-side bus stop, Bus stop at a lay-by (12)
and Bus cape with necessary sketches.
- (b) What are the factors that influence the required bus fleet strength of a city? (12)
Calculate the required fleet strength of buses for Khulna city from the following data set:
Population: 1,150,341; Per Capita Trip Rate: 1.7;
Share of bus trips in overall trips: 25%; Average trip length: 8.2 Km;
Distance travelled by bus per day: 250 km; Capacity of Bus: 70 passengers;
Average occupancy: 0.8; Expected fleet utilization: 95%.
- (c) How city bus routes can be classified? Mention the needs and functions of Bus (11)
Route Planning.
3. (a) ABCD is a street running North to South comprising of 3 adjacent sections. (17)
Table 01 below gives the length of each section:

Table 01

Section	Length (m)
AB	500
B	Intersection
BC	600
C	Intersection
CD	450

Table 02 and 03 give the details of observations (average of the test runs) by the moving observer team:

Calculate the flows and running speeds in each section in both directions and the overall journey speeds in each direction.

Table 02 (Travelling South)

Section	Time taken to traverse (sec.)	Vehicles met with in opposing direction	Vehicles in the same direction	
			Vehicles overtaking	Vehicles overtaken
AB	152	25	4	3
B	15	-	-	-
BC	172	31	2	2
C	10	-	-	-
CD	138	28	3	4

Table 03 (Travelling North)

Section	Time taken to traverse (sec.)	Vehicles met with in opposing direction	Vehicles in the same direction	
			Vehicles overtaking	Vehicles overtaken
DC	132	50	2	1
C	12	-	-	-
CB	160	55	3	4
B	18	-	-	-
BA	145	45	1	1

- (b) It is observed that on an average a vehicle driver drives 5,000 km during the course of a year. The probability of having an accident is 100 per 200 million vehicle-kilometers. What is the probability of a driver having at least 3 accidents during his driving career extending to 25 years? (10)
- (c) The accident data pertaining to a metropolitan city for the year 1965 and 1970 are given below: (08)

	1965	1970
Accidents	300	400
vehicle-kilometer of travel	250 million	300 million

Test whether there is any significant increase in the accident rates in two years. [The critical value of χ^2 is 3.841 with 5% level of significance].

4. (a) Using simple queuing theory derive: $P_n = \rho^n(1 - \rho)$; where ρ means traffic intensity, n means number of items and P_n means possibility that the queuing system has n items at a given time. (The service pattern is exponentially distributed and the flow has a poissonian arrival pattern). (20)
- (b) A toll booth at the entrance to a bridge can handle 120 veh/hour, the time to process a vehicle being exponentially distributed. The flow is 90 veh/hour with a poissonian arrival pattern. Determine: (08)
- the average number of vehicles in the system;
 - the length of the queue;
 - the average time spent by the vehicle in the system;
 - the average time spent by the vehicle in the queue.
- (c) "The growth of a new mode of transportation favours the decline of others". (07)
Explain this sentence in light of transport and economic opportunities.

Section – B

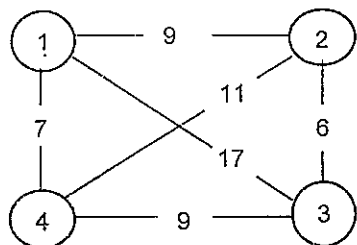
5. (a) What are the basic elements of transportation planning? Explain those with appropriate examples. (13)
- (b) What is Traffic Analysis Zone (TAZ)? Mention the selection criteria of TAZ. (07)
- (c) Let the number of trips from i to j is 15,000. Three modes are available from i to j to make the trips which has the characteristics given in the Table. Compute the trips carried by the three modes and the fare required to travel by each mode. (15)

	Vehicle travel time between 'i' and 'j'	Walking time to and from stops	Waiting time at stops	Fare charged to travel between 'i' and 'j'	Parking cost
Co-efficient	0.03	0.04	0.06	0.1	0.1
Car	25	-	-	15	3
Bus	35	8	4	8	-
Train	17	12	3	5	-

6. (a) Write a short note on Trip Matrix with necessary examples. (05)
- (b) Travel mode choice depends on a number of factors. Explain those factors with examples. (10)
- (c) What do you understand by aggregate and disaggregate level of analysis in transportation planning? (05)
- (d) A residential zone produces a total of 190 shopping trips per day. Distribute these trips to shopping centers 1, 2, 3 and 4 in according to the gravity model. Distances between residential area to shopping centers 1, 2, 3 and 4 are 6, 8, 4 and 5 miles respectively. Use commercial floor space within the destination as the measure of attractiveness. (15)

Shopping center	Floor space (thousand sq. ft)
1	184
2	215
3	86
4	105

7. (a) What are the advantages and disadvantages of All-or-Nothing Assignment? (10)
- (b) "Trip assignment is based on two assumptions of traveler's behavior". Explain those with examples. (08)
- (c) A highway network consisting of four nodes. A trip table (veh/hr) needs to be loaded on the network. Calculate the new link travel time using capacity restrained assignment. Assume capacity of each link to be 650 veh/hr. (17)



To From	1	2	3	4
1	0	350	550	210
2	220	0	450	590
3	530	800	0	400
4	410	500	620	0

8. (a) Write short note on the following TDM measures with examples: (12)
(i) Vehicle Restrictions; (ii) Alternative Work Schedules.
- (b) What TDM measures will you recommend for Dhaka city to manage travel demand of the people? (10)
- (c) "Total motorization demonstrates modernization of urban transportation in a country". Do you agree with the statement? Explain your understanding with examples. (05)
- (d) Do you think Intermodal Coordination is an effective measures of travel demand management? Give a logical explanation with appropriate examples. (08)

Khulna University of Engineering & Technology
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BURP 3rd Year 2nd Term Regular Examination, 2016
URP 3223
(Regional Development Planning)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is your understanding about different types of region? Explain with examples. (10)
- (b) How will you delineate a formal region using weighted index number method? Explain with appropriate examples. (15)
- (c) Mention different types of multiplier and their contribution to regional development. (10)
2. (a) What are the principles that Fawcett followed to divide England into 12 provinces? (10)
- (b) Find the gravitational potential of a centre A from the following information, where A is surrounded by the centers B, C & D. (07)

Centre	Employment	Distance from A (m)
B	2000	200
C	2500	500
D	1000	250

Here, the value of $k = 1.75$

- (c) How does shift-share component contribute to industrial structure analysis? Explain in short. (12)
- (d) What are the limitations of economic base theory? Explain in short. (06)
3. (a) Exemplify the indirect approaches used for classifying the basic and non-basic activities. (15)
- (b) The following table shows different industry's employment of a region for nation along with national growth rate (13.6%), industry sector's national growth rate and regional industry's growth rate. (20)

Industry	Regional Employment (1993)	Industry's National Growth Rate (%)	Regional Industry's Growth Rate (%)
Agriculture & Fishing	79691	-0.30	0.40
Manufacturing	67643	2.84	3.65
Mining	53268	15.01	13.25
Finance & Real Estate	72523	15.75	12.65
Service	38622	18.46	20.70
All others	78476	11.78	13.90

- (i) Calculate National Growth Share (NS), Regional Growth Shift (RS) and Industry Mix (IM) separately for each industry.
- (ii) Find out the fast growing and declining industries.
4. (a) What do you mean by aggregated and disaggregated growth models? What are the basic differences between them? (08)
- (b) The South-East economy was dependent on two industries only: A, employing 5 million people at the beginning of the periods and B, employing 3 million and both were growing faster than the national rate of approximately 6%, A at 8% and B at 11%, find out the shift component. (10)

- (c) Mention different levels of planning. Exemplify the role of inter and intra regional planning in regional development. (10)
- (d) Write short note on sub-regional level planning. (07)

Section – B

5. (a) "Appropriate regional data base development is vital for a developing country like Bangladesh having heterogeneous spatial and economic characteristics"-why? Explain with proper example. (10)
- (b) Make a comparison between endogenous and exogenous determinants of regional growth. (05)
- (c) Discuss about any two regional growth models determined by the endogenous and exogenous factors of regional growth. (10)
- (d) Write short notes on any two of the following issues: (10)
- (i) Interrelationship among the factors of production in production-function type growth theory
 - (ii) Inter-regional resource allocation model
 - (iii) Export base theory
6. (a) What is your understanding about regional spatial structure? Specify any two theories indicating spatial structure of any region. (05)
- (b) Make a comparison between the key features of "Central Place Theory" and "Growth Pole Theory" with necessary diagrams. Which one is more applicable for Bangladesh? Why? (12)
- (c) How "Polarization" is explained in "Growth Pole Theory"? Discuss about the agglomeration economies identified in "Growth Pole Theory" with necessary example. (10)
- (d) Write short notes on the following issues: (08)
- (i) Rank Size Rule
 - (ii) Gravity Model
7. (a) Briefly discuss the "Umbrella Approach" of industrial location theory using Smith's diagram of average cost and revenue. (10)
- (b) Locational choice of any industry is based on three major considerations according to "Least Cost Approach"- explain using necessary diagram. (15)
- (c) Make a list of influential factors for selecting a new effective location of any industry. (10)
8. (a) Write about two analytical models demarcating economic structure of any region. (05)
- (b) "Four sections of input-output table represent four different relations of industrial transaction"- explain these relations using an example. (08)
- (c) According to the input coefficient matrix find out the total input requirements for the final demand 3000 in manufacturing sector and 1000 in commercial sector (after 2nd round calculation). (15)

Output Input	Manufacturing	Commercial	Household
Manufacturing	0.50	0.20	0.50
Commercial	0.33	0.30	0.38
Household	0.17	0.50	0.13

- (d) Make a comparison between "Transaction Matrix" and "Input Coefficient Matrix" in regional input-output analysis. Use an example. (07)

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Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2016

URP 3291
(Operations Research and Systems Analysis)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary chart and graphs will be supplied.



Section – A

1. (a) What is "Operations Research"? Briefly explain different fields of Operations Research. (13)
(b) Write down the phases of OR study. (07)
(c) What is meant by "Modelling & Simulation"? Give some real life application examples of M & S. (15)
2. (a) A firm makes two types of furniture - chairs and tables. The contribution for each product as calculated by the accounting department is Rs. 20 per chair and Rs. 30 per table. Both products are processed on three machines M_1 , M_2 and M_3 . The time required (in hours) by each product and total time available per week on each machine are as follows: (15)

Machine	Chair	Table	Available hours per week
M_1	3	3	36
M_2	5	2	50
M_3	2	6	60

How should the manufacturer schedule his production in order to maximize contribution? Formulate the problem as a linear programming problem.

- (b) Solve the following linear programming problem by graphical method: (15)

Maximize, $z = 5x_1 + 4x_2$
Subject to, $6x_1 + 4x_2 \leq 24$
 $x_1 + 2x_2 \leq 6$
 $-x_1 + x_2 \leq 1$
 $x_2 \leq 2$
 $x_1, x_2 \geq 0$

- (c) What is meant by "Shadow Price"? (05)

3. (a) What is Gauss-Jordan Row Operations? (07)

- (b) Solve the following LP problem using Simplex Method: (18)

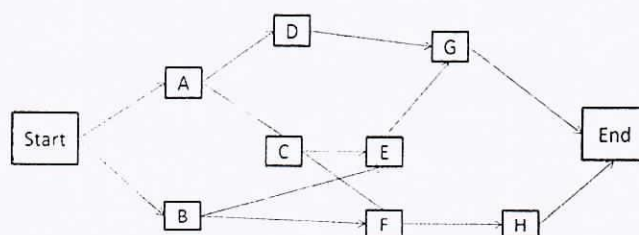
Maximize, $z = x_1 + 4x_2 + 5x_3$
Subject to, $3x_1 + 3x_3 \leq 22$
 $x_1 + 2x_2 + 3x_3 \leq 14$
 $3x_1 + 2x_2 \leq 14$
 $x_1, x_2, x_3 \geq 0$

- (c) Obtain the dual of the following linear programming problem: (10)

Maximize, $z = 20x_1 + 50x_2$
Subject to, $5x_1 + 4x_2 \leq 20$
 $3x_1 + 5x_2 \leq 15$
 $x_1, x_2 \geq 0$

4. (a) Discuss different types of simulation process with necessary examples. (15)

- (b) Given the estimated activity times and the network below, (20)



Activity	a	m	b
A	6.5	7.5	14.5
B	8.5	10.5	12.5
C	2.5	3.5	4.5
D	6.5	7.5	8.5
E	5.5	5.5	9.5
F	5.5	7.5	9.5
G	4.5	6.5	8.5
H	2.5	3.5	3.5

What is the critical path probability that the project will be completed within:
(i) 21 days? (ii) 22 days? (iii) 25 days?

Section – B

5. (a) Consider the following payoff (profit) matrix. (16)

	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅
a ₁	15	10	0	-6	17
a ₂	3	14	8	9	2
a ₃	1	5	14	20	-3
a ₄	7	19	10	2	0

No probabilities are known for the occurrence of the future states. Compare the solutions obtained by the following criteria:

(i) Laplace, (ii) Minimax, (iii) Savage, (iv) Hurwicz (Assume that $\alpha = 0.5$)

- (b) A person has two independent investments A and B available to him, but he can undertake only one at a time due to certain constraints. He can choose A first and then stop, or if A is successful then take B or vice versa. The probability of success of A is 0.6, while for B it is 0.4. Both the investments require an initial capital outlay of BDT 10,000 and both return nothing if the venture is unsuccessful. Successful completion of A will return BDT 20,000 (over cost) and successful completion of B will return BDT 24,000 (over cost). Draw decision tree and determine the best strategy. (19)

6. (a) Solve the following game graphically. (23)

		B	
		1	2
A	5	5	6
	-7	-7	9
	-4	-4	-3
	2	2	1

- (b) Define "Queuing System". Briefly explain the elements of a Queuing System using 'a barber shop' as example. (12)

7. (a) Patients arrive at a clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 14 patients. Examination time per patient is exponential with mean rate 20 per hour. (15)

(i) Find the effective arrival rate at the clinic.

(ii) What is the probability that an arriving patient will not wait?

(iii) What is the probability that an arriving patient will find a seat?

- (b) A telephone exchange has two long distance operators. It is observe that, during the peak load, long distance calls arrive in a Poisson fashion at an average rate of 15 per hour. The length of service on these calls is approximately exponentially distributed with mean length of 5 minutes. (20)

(i) What is the probability that a subscriber will have to wait for his long distance call during the peak hours of the day?

(ii) If the subscribers will wait and are serviced in turn, what is the expected waiting time?

8. (a) Differentiate between "Decision Under Risk" and "Decision Under Uncertainty" with suitable example. (10)

- (b) What is "Decision Tree"? Write down the basic rules and conventions for drawing decision tree. (15)

- (c) List the software used for modelling and simulation purpose by an Urban Planner. (10)